

MediTrace OS (KIT26038)

Smart Bedside Medical-Waste Collection, AI Computer Vision & IoT Digital Twin OS

A next-generation healthcare platform retrofitted onto hospital ward trolleys to eliminate bedside cross-contamination, enforce India CPCB BMWM Rules 2016 statutory compliance, and automate digital audit manifest generation.

AI Computer Vision

Gemini Vision AI classifying waste into 4 statutory bins with Trilingual Vernacular Voice Directives (English, Hindi, Telugu).

IoT Solenoid Interlocks

Dynamic solenoid lid locks (UNLOCKED target bin, LOCKED non-target bins) & 880Hz Web Audio emergency siren.

CPCB Form-IV Manifest

Automated shift sealing with itemized category weights and signed 2D serialization QR codes for CBWTF pickup.

The Hospital Bedside Cross-Contamination Crisis

CHALLENGE & OPPORTUNITY

& p The Clinical & Regulatory Problem

- Heavy Statutory Penalties: BMWM Rules 2016 penalize healthcare facilities for improper bedside segregation.
- Pathogen Exposure Risk: Ward nurses and cleaning staff face extreme infection risks from mis-binned anatomical waste and sharps.
- Paper Log Mismatches: Manual paper logs lead to disputes between hospital waste managers and CBWTF transport truck operators during pickup.

Ø=Ü; The MediTrace OS Solution

- Retrofit Intelligence: Mounts lightweight camera, AI vision scanner, and solenoid locks onto standard hospital ward trolleys.
- Automated Lid Interlocks: Only the verified target bin lid unlocks; non-target lids remain locked to physically block missegregation.
- Immutable Digital Manifests: Every disposal logs weight telemetry, generates signed Form-IV QR codes, and syncs with CBWTF transport fleets.

Multilingual AI Vision Engine & Vernacular Directives

Ø=ßá YELLOW CONTAINER

- Infectious anatomical waste, soiled cotton, adult diapers.
- Protocol: 1050°C Incineration / Pyrolysis
 - Recyclability: 0% (Extreme Pathogen Risk)

Ø=Ý4 RED CONTAINER

- Contaminated plastics: IV drip tubing, saline bottles, catheters.
- Protocol: Autoclave Sterilization & Shredding
 - Recyclability: 95% Recyclable Plastics

Ø=Ý5 BLUE CONTAINER

- Glassware: Medicine vials, ampoules, metallic implants.
- Protocol: Sodium Hypochlorite Disinfection & Smelting
 - Recyclability: 85% Glass Recycling Yield

&ª WHITE TRANSLUCENT

- Sharps: Syringe needles, surgical scalpels, lancets, blades.
- Protocol: Autoclaving & Needle Mutilation
 - Recyclability: 10% Encapsulated Scrap

Ø=Ýãþ Trilingual Vernacular Voice Directives Engine (EN / HI / TE)

- Integrated Web Speech API (te-IN, hi-IN, en-US locales) providing real-time spoken vernacular instructions for frontline hospital ward staff:
- English: "Drop IV tubing into Red recyclable plastics bin."
 - Hindi ("™?•)&" " "y 'iG 'YI'ù('lO ' >•y*ù "2 *" 2"r 8•) "Ü0"é: 'R !"ù, "Ü, "r . "y ' >")G•)d"
 - Telugu (ÂLFÄ,AÁ|A): "ÄÆBÄI? Äi0Äü/Ä 8ÄÜ Ä|2ÄÜ*Äl2ÄÖ Î(Ä (Ä|0Ä Äâ \$Äl2Ä *Ä 0Ä, Ä *Ä, ÄÜ Ä M-Ä¬MÄ BÄ¼M Ä 8ÄÜ ÄÖ ,Äü(ÄÖ Î2Ä² 5Ä|/Ä,Äðâ

Complete Architectural Technology Stack

Layer / Category	Technologies & Libraries	Architectural Purpose & Core Implementation
Frontend Framework	Next.js 14.2 (App Router), React 18	Server-side rendering, client components & serverless API routes
Type System & Language	TypeScript 5.4	Strict type safety across telemetry, manifests & classification schemas
Styling & Design System	Tailwind CSS 3.4, Custom Glassmorphism	Responsive 2-column grid, red emergency flash keyframes & dark medical UI
AI Vision Engine	Gemini 1.5 / 2.0 Flash Vision API	@google/generative-ai serverless image analyzer with BMW prompt
Speech & Audio Engine	Web Speech API & Web Audio API	Trilingual TTS (EN, HI, TE), 880Hz Siren Oscillator & High C Scan Chime
IoT Telemetry & Solenoids	Load Cell Telemetry State Machine	Weight tracking (kg), fill level bars (%), Tare zero reset calibration
Barcode Serialization	qrcode (1.5.3) & canvas-confetti	Signed 2D QR serialization canvas for CBWTF vehicle handover
Logistics Fleet Portal	HTML5 Route Canvas & Reconciler	Weighbridge scale delta calculator & live 14.2km GPS transit map

CPCB Form-IV Manifest & CBWTF Logistics Portal

Ø=ÜÄ CPCB Form-IV Digital Manifest Modal

- Automated Shift Batch Sealing: Triggered via "Seal Batch / Form-IV" button with celebratory confetti.
- Itemized Category Weight Table: Displays total shift payload (kg) and category breakdown for Yellow, Red, Blue, and White bins.
- 2D Serialization QR Code: Generates a signed QR code canvas containing manifest payload and SHA verification hash (0x94f8a...e102c9).

Ø=Þ> CBWTF Transport Fleet & Weighbridge Portal

- QR Manifest Reader: CBWTF truck driver scans ward Form-IV QR code to import bedside payload metrics.
- Weighbridge Scale Reconciler: Compares Bedside logged weight vs Truck Loading Dock scale and flags 9F¶r F—67&W æ7' ÆW ts.
- Live GPS Transit Map: Simulates vehicle route from Kasturba Hospital -> CBWTF Incineration Plant (14.2 km).
- Digital Handover Receipts: Dual signature sign-off issuing sealed Transfer Certificate (#CBWTF-RC-9941).

System Verification Audit & Production Readiness

Automated Browser Subagent Verification Audit (100% Pass)

Test Scenario	Target Component	Observed Behavior	Status
1. Preset Classification	CameraScanner.tsx	Correct BMW bin returned	PASS
2. Trilingual Speech	SpeechSynthesis	EN, HI, TE directives spoken	PASS
3. Solenoid Interlocks	CartDigitalTwin.tsx	Target UNLOCKED; 3 LOCKED	PASS
4. Mismatch Siren	Web Audio Oscillator	Red flash & 880Hz siren active	PASS
5. Tare Load Cell Reset	Cart Header	All 4 bins reset to 0.00 kg	PASS
6. Form-IV Manifest QR	CPCBManifestModal	Confetti & signed QR rendered	PASS
7. Logistics Handover	/logistics Route	Weight delta & GPS map synced	PASS

Build Status: npx tsc --noEmit PASSED (0 Errors) • npx next build PASSED (Exit Code 0)

Strategic Expansion Roadmap

- MediTrace ColdChain: Real-time BLE temperature telemetry for vaccine & blood bank excursions.
- MediTrace RxGuard: AI OCR + GS1 2D barcode scanner for anti-counterfeit drug verification.
- ESP32 Hardware Wiring: MQTT / WebSocket bridge connecting physical HX711 load cells and 12V solenoid locks.